

Understanding Cyclical Businesses

Market psychology, structural mechanics, and mental models for investors

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This note compiles the ideas discussed across a conversation on why businesses and markets move in cycles — covering both sentiment-driven bubbles and structurally mechanical cycles — with real historical examples, worked numerical illustrations, and a set of transferable mental models for evaluating cyclical industries.

1. The Business Cycle: Four Phases, Four Contradictions

A business cycle isn't a flaw in the system — it's driven by the same forces that create growth, just running past their useful point. Four contradictions sit at each turning point of the cycle:

Phase	The contradiction	Signs to watch for
Expansion	Confidence is the fuel and the flaw — growth's success quietly builds the excess that ends it.	Employment & wages rising; credit expanding faster than income; profits and asset prices climbing broadly; low rates, wide optimism.
Peak	Stability breeds instability (Minsky) — calm periods encourage the risk-taking that later makes the system fragile.	"This time is different" talk; valuations far above norms; leverage accelerating; central banks raising rates; IPO/new-entrant frenzy.
Contraction	The crowd's exit causes the damage it fears — panic selling overshoots fundamentals.	Credit tightening; rising defaults; broad forced liquidations; layoffs begin; spending slows.
Trough	The worst-feeling moment is often the best-timed one — total pessimism usually means prices overshot down.	Capitulation selling; valuations below long-run averages; aggressive rate cuts; bankruptcies cluster, then slow.

2. Five Historical Bubbles, Five Sectors, Centuries Apart

Era	Place	Sector	What happened
1634–1637	Dutch Republic	Commodities (tulip bulbs)	Contract prices for tulip bulbs reached extraordinary levels, then collapsed within days after a Haarlem auction found no buyers in Feb 1637 — the first recorded speculative bubble.
1986–1991	Japan	Real estate & equities	The Nikkei peaked near 38,916 in Dec 1989; commercial land prices then fell for roughly 14 straight years, feeding Japan's "lost decade."
1995–2002	United States (global)	Technology / equities	The Nasdaq rose 572% from 1995 to its March 2000 peak of 5,048, then fell 78% by October 2002.
2002–2009	United States (global)	Housing / banking	A subprime lending boom collapsed into mass foreclosures; Lehman Brothers' Sept 2008 bankruptcy (the largest in U.S. history) triggered a global financial crisis.

Era	Place	Sector	What happened
2020–present	China	Real estate development	After debt limits hit developers in 2020, Evergrande defaulted in Dec 2021; companies representing about 40% of Chinese home sales have since defaulted too.

The recurring pattern: the crisis always arrives disguised as something new. Tulips, Nikkei land prices, dot-com IPOs, mortgage securities, Chinese property debt — different assets, same shape.

3. Two Kinds of Cyclicity: Psychology vs. Structure

Tulip mania and 2008 needed crowd psychology — greed, fear, “this time is different.” But industries like cement, banking, and semiconductors would cycle even if every person in them behaved perfectly rationally. That is the deeper contradiction: remove all the psychology, and the cycle survives — because it's built into the physics of the business, not the emotions of the people running it.

3.1 Capacity Lag — The Cobweb Theorem

First formalized from 1920s German hog markets (Hanau, 1928) and named the cobweb theorem by Nicholas Kaldor in 1934: producers expand supply based on today's price, but new supply only arrives after a fixed production lag — by which point conditions have often reversed. No irrationality required; the cycle emerges purely from the lag between decision and delivery.

- Cement: pricing power only returns once capacity utilisation clears roughly 70–75%; India's sector is adding 180–200 million tonnes of capacity for FY26–FY28, which analysts expect will keep utilisation below that threshold for years.
- Semiconductors: a new fab takes 18–36 months from investment decision to output. The industry has logged more than 15 downcycles in its 70-year history — the 2023 downturn alone cut memory sales by 31%.
- Housing: researchers have found the same cobweb pattern in the Irish and Israeli housing markets — supply surges just as the boom that justified it is ending.

3.2 Leverage & Recognition Lag — Banking

A bank's profits look best exactly when its future losses are being created. Loose lending in an expansion shows up as growth today and defaults years later, after borrowers hit their own downturn. The cycle in banking is inherited, with a lag, from every borrower's cycle at once.

- India's gross NPA ratio rose from 2.4% (March 2011) to 11.5% (March 2018) — the delayed bill for lending done in the 2003–2008 boom, much of it only surfaced after the RBI's 2015 Asset Quality Review forced recognition.
- By September 2025 the system-wide gross NPA ratio had fallen to 2.15%, aided by roughly ■14.6 lakh crore in loan write-offs between FY15 and FY23.

- This had happened before: public sector bank NPAs hit 25% in 1994 after the lending boom of the prior decade — same mechanism, one cycle earlier.

3.3 Distance from the End Consumer — The Bullwhip Effect

A small, real change in end-consumer demand gets amplified at every step backward through the supply chain, because each layer over-orders on the way up (adding its own safety buffer) and over-cancels on the way down. A worked example:

Step in the chain	Change in orders
Consumer buys phones	+5%
Retailer orders from phone maker	+10%
Phone maker orders chips	+20%
Chipmaker orders fab equipment	+40%

Why: (1) buffers stack — each layer pads its own order on top of the layer before it as a safety margin; (2) nobody sees the real signal, only the order in front of them — the equipment maker sees a 20% jump in chip orders and reads it as a much bigger boom than the 5% rise in real consumer demand that started it. It runs in reverse on the way down, too: a small real slowdown causes retailers to stop ordering, which causes equipment makers to see almost nothing coming in.

- Semiconductor equipment makers have historically swung about -20% per downcycle versus roughly -8% for the chips themselves.
- This is called being on the economy's 'second derivative': equipment cycles are driven not by GDP falling, but by GDP's rate of growth merely slowing.
- Practical rule: the further back a business sits from the end consumer in its value chain, the more violent its cycle will be — independent of how good or bad the business itself is.

4. The P/E Paradox in Cyclical

Normally, low P/E = cheap and high P/E = expensive. In cyclicals, this can run exactly backwards, because the E (earnings) itself swings wildly across the cycle. A simplified, realistic-shaped example for a cement company:

Year	What's happening	EPS (■)	Price (■)	P/E
2020	Trough — demand weak, prices depressed	5	100	20x
2021	Recovery starting	10	150	15x
2022	Boom — high utilisation, strong pricing power	20	200	10x
2023	Peak — everyone expanding capacity	25	200	8x
2024	Oversupply hits, prices crack	12	150	12.5x
2025	Bust — earnings collapse	5	90	18x

In 2023 the stock looks cheapest ever at 8x earnings — but that EPS is a cyclical peak about to mean-revert down; buying “cheap” here means buying right before earnings halve. In 2020 or 2025 the stock looks expensive at 18–20x — but that EPS is a depressed, cyclically-low number; the same price becomes very cheap once earnings recover. **Fix:** don't trust the P/E on trailing or current-year earnings for a cyclical. Either judge where the business sits in its capacity-utilisation cycle, or value it against a normalized “mid-cycle” earnings figure averaged across a full boom-and-bust.

5. Four More Mental Models

5.1 Operating Leverage

When most costs are fixed (leases, crew, plant maintenance) rather than variable, revenue changes flow almost entirely into profit — in both directions.

- Delta Air Lines swung from a \$4.8 billion profit in 2019 to a \$12.4 billion loss in 2020 — revenue fell hard, but leases, aircraft, and crew costs stayed largely fixed.
- Rule of thumb: high fixed-cost businesses can see a 10% revenue rise become a 30% profit jump, and a 10% revenue decline trigger a 30% profit collapse.
- Before buying any cyclical, ask what share of its cost base is fixed — that number tells you how violent the earnings ride will be, independent of the industry's demand cycle.

5.2 The Cost Curve

Line up every producer in an industry from cheapest to most expensive cost per unit. The market price settles at the cost of the marginal, highest-cost producer still needed to meet total demand — because that producer won't sell below its own cost, and every cheaper producer below it earns the gap between the

market price and its own (lower) cost as pure profit.

Worked example — six copper mines, demand = 16 Mt:

Mine	Cost (\$/t)	Capacity (Mt)	Cumulative (Mt)	Status at 16 Mt demand
A	3,000	5	5	Profitable (+\$5,000/t)
B	4,500	4	9	Profitable (+\$3,500/t)
C	5,500	4	13	Profitable (+\$2,500/t)
D	7,000	3	16	Marginal — sets the \$7,000 price
E	8,000	4	20	Idle — cost exceeds price
F	9,500	3	23	Idle — cost exceeds price

If demand rises, a costlier mine becomes marginal and the price steps up. If demand falls, a cheaper mine becomes marginal and the price steps down — and whichever mine was previously marginal now goes idle. This is the literal mechanism behind “low-cost producers survive downturns”: it isn’t a strategic slogan, it’s a direct result of where a company sits on this ladder. The same logic applies to cement plants, airline fleets, and even bank funding costs — anywhere competitors can be ranked by cost.

Real-world evidence: thermal coal traded below the industry’s 90th-percentile cost level for three straight years before loss-making mines finally began mothballing capacity — the cost curve can be violated for a long time before it corrects.

5.3 Capital Cycle Theory

Associated with Edward Chancellor and Marathon Asset Management: most investors obsess over demand forecasts, which are notoriously hard to get right. Capital cycle theory instead watches supply — how much capital and capacity is flooding into an industry right now. Heavy investment today predictably erodes returns years later when that capacity arrives; capital fleeing an industry predictably sets up the next recovery, almost regardless of the demand story.

- Core finding from two decades of Marathon’s portfolio reports: industry returns mean-revert primarily because of changes in the rate of capital expenditure, not changes in demand.
- Practical rule: avoid sectors where investment is elevated and competition is intensifying; look for sectors where capital has been scarce for years with few new entrants.
- This is the investing strategy built directly on top of the capacity-lag / cobweb mechanism described in Section 3.1.

5.4 Structural vs. Cyclical Decline

A cyclical downturn is temporary — demand is delayed, not destroyed, because the underlying need doesn’t disappear. A structural decline is permanent — the product itself is being replaced. Confusing the two is arguably the costliest mistake in cyclical investing.

- Kodak: film held roughly 90% U.S. market share in the 1970s; digital photography made it essentially obsolete within about a decade of digital cameras going mainstream, and Kodak filed for bankruptcy in 2012. No cyclical recovery was ever coming — the need itself (chemical film) had been substituted away.
- Contrast with cement or autos: a recession delays construction and car purchases, but buildings still age and cars still wear out — deferred demand doesn't vanish, it queues up and often returns as a sharper-than-normal recovery.
- The diagnostic question: has the underlying human need disappeared or been substituted by something structurally better/cheaper — or has spending on an unchanged need simply been postponed by the credit or economic cycle?

6. Industrial Headwinds & Tailwinds Framework

Every force acting on an industry falls into one of four boxes — and the market frequently misprices which box a given force belongs in.

	Tailwind (helps)	Headwind (hurts)
Structural (permanent, doesn't reverse)	Urbanization, electrification, rising incomes, an aging population needing more healthcare	Technological substitution (film → digital), demographic decline, resource depletion, a product regulated out of existence
Cyclical (temporary, mean-reverts)	Falling interest rates, an inventory restocking phase, a commodity price spike	Rising rates, an inventory destocking phase, a recession-driven pause in discretionary spending

Two ways this grid gets mispriced, both expensive: mistaking structural decline for a cyclical dip (buying “cheap” Kodak because “photography always recovers”, when the category itself was being permanently substituted away); and mistaking a cyclical trough for structural decline (selling cement or auto stocks in a recession because “this industry is dying,” when the need was only deferred).

6.1 Cycles Are Nested Inside Each Other

What looks like “the cycle turning” may just be one small cycle inside a much bigger one still running the other way. Economists have long described overlapping cycles at different speeds:

- A short inventory cycle (roughly 2–4 years) — the destocking/restocking rhythm behind the bullwhip effect.
- A medium capex/investment cycle (roughly 7–11 years) — the capacity-lag cycle cement, semiconductors, and shipping run on.
- A long infrastructure/building cycle (roughly 15–25 years) — Japan's 1986–1991 bubble and its multi-decade unwind is a textbook example.
- A very long structural/technology wave (decades) — steam, rail, electrification, internet, and now AI, each reshaping which industries even exist.

Practical implication: an industry can be in a cyclical upswing (short cycle recovering) while sitting inside a structural headwind (a long-run trend ending), or vice versa. When two readings of “the cycle” seem to contradict each other, it's often because each is looking at a different nested layer, not because one is wrong.

7. Practical Investor Checklist

- Ask how many links from the end consumer a business sits — distance amplifies volatility (the bullwhip effect).

- Watch capacity utilisation as the leading indicator, not trailing earnings — know the sector's pricing-power threshold (e.g. ~70–75% for cement).
- Flip your read of the P/E ratio in cyclicals — a low multiple at peak earnings can be a trap; a high multiple at trough earnings can be a bargain. Normalize to mid-cycle earnings.
- Know the fixed-cost share of a business — it sets how violently profit will swing relative to revenue (operating leverage).
- Rank competitors on the cost curve — a company's cost position, not its growth rate, often decides whether a downturn is survivable.
- Track capital flowing into an industry, not demand forecasts — heavy investment today predictably erodes returns years later (capital cycle theory).
- In leveraged cyclicals (banks, developers), prioritize balance-sheet survivability over growth — the winners are the ones still standing at the next upcycle.
- Diagnose structural vs. cyclical before anything else: has the underlying need disappeared/been substituted, or just been postponed?
- Remember cycles nest — a short-cycle recovery can sit inside a long-cycle or structural headwind, and vice versa.

Compiled from a conversation with Claude (Anthropic). Historical figures and statistics were sourced via web search during the conversation; treat this as a study reference rather than investment advice.